Answer Key to Problem set 2

Econ 5030

1 Problem 1

The problem is based on the game matrix below.

		Phil		
		n_P	l_P	h_P
	n_G	$0,0$	$1,2$	$1,2$
Greg	l_G	$3,1$	$3,3$	$3,4$
	h_G	$2,1$	$4,3$	$5,5$

1.A). Find the solution set of strategies based on iterated deletion of strictly dominated strategies.

Answer:

Deletion 1

		Phil		
		n_P	l_P	h_P
	n_G	$-$	$-$	$-$
Greg	l_G	$3,1$	$3,3$	$3,4$
	h_G	$2,1$	$4,3$	$5,5$

Deletion 2

		Phil		
		n_P	l_P	h_P
	n_G	$-$	$-$	$-$
Greg	l_G	$-$	$3,3$	$3,4$
	h_G	$-$	$4,3$	$5,5$

Deletion 3

		Phil		
		n_P	l_P	h_P
	n_G	$-$	$-$	$-$
Greg	l_G	$-$	$-$	$3,4$
	h_G	$-$	$-$	$5,5$

Deletion 4

		Phil		
		n_P	l_P	h_P
	n_G	$-$	$-$	$-$
Greg	l_G	$-$	$-$	$-$
	h_G	$-$	$-$	$5,5$

1.B). Find the set of Nash equilibrium.

Answer: (h_G, h_P)

2 Problem 2

Solving the prisoner's dilemma

Normal Form of the Game (Game matrix):

		Bob	
		q_B	f_B
Al	q_A	$2,2$	$0,3$
	f_A	$3,0$	$1,1$

2.A). Find the set of Nash equilibrium.

Answer: (f_A, f_B)

3 Putting games into the Normal Form

Consider the following 2 stage game $t = 1, 2$. The game has 2 players. In stage 1 player 1 takes one of two $\{a_1, b_1\}$. In stage 2 player two first

observes that action of player 1 in stage 1 and then takes one of three actions $\{c_2, d_2, e_2\}$. The payoff of each player is as follows.

$$\begin{array}{cc} u_1(a_1, c_2) = 10 & u_2(a_1, c_2) = 8 \\ u_1(a_1, d_2) = 8 & u_2(a_1, d_2) = 9 \\ u_1(a_1, e_2) = 12 & u_2(a_1, e_2) = -11 \\ u_1(b_1, c_2) = 13 & u_2(b_1, c_2) = 0 \\ u_1(b_1, d_2) = 1 & u_2(b_1, d_2) = 10 \\ u_1(b_1, e_2) = 10 & u_2(b_1, e_2) = 6 \end{array}$$

1. Draw the game tree of the game.

2. Write out the set of strategies for each player.

   Answer: Player 2 Strategies:

   $$\{a_1, b_1\}.$$

   Player 1's strategies

   $$\{c_2c_2, c_2d_2, c_2e_2, d_2c_2, d_2d_2, d_2e_2, e_2c_2, e_2d_2, e_2e_2\}$$

3. Draw the normal form matrix for the game.

   Answer:

	a_1	b_1
c_2c_2	$10, 8$	$13, 0$
c_2d_2	$10, 8$	$1, 10$
c_2e_2	$10, 8$	$10, 6$
d_2c_2	$8, 9$	$13, 0$
d_2d_2	$8, 9$	$1, 10$
d_2e_2	$8, 9$	$10, 6$
e_2c_2	$12, -11$	$13, 0$
e_2d_2	$12, -11$	$1, 10$
e_2e_2	$12, -11$	$10, 6$

4. Find all Nash equilibrium.

   Set of NE

   $$b \text{ ec}$$

5. Are there equilibrium that are based on incredible threats? Explain

   Answer: Only (b, ec) is subgame perfect.

4 Timing and Strategies

Take a simple two player game with each player having actions 0 or 1. Player 1 gets \$1 if the actions match and \$0 if they do not match. Player 2 gets \$0 if the actions match and \$1 if they do not match.

2.1 Suppose the game is simultaneous choice game. Explain the strategies of each player.

Answer: $S_i = \{0, 1\}$

2.2 Suppose the game is sequential choice game in which player 1 goes first, then player 2 observes player 1's choice and makes her choice. Explain the strategies of each player.

Answer: $S_1 = \{0, 1\}$, $S_2 = \{00, 01, 10, 11\}$

2.3 Explain the difference in the normal form of these two games.

Answer: only difference is strategies

5 Third party observation of a game

Suppose that you run an experiment in a laboratory with real money in which there are many pairs of players and the game has the following structure.

Each player i has three actions $\{L_i, M_i, H_i\}$. The money each player i receives for the different action pairs is (in a game with player j) is described in the matrix below:

	L_1	M_1	H_1
L_2	\$8, \$8	\$30, \$10	\$15, \$1
M_2	\$10, \$30	\$18, \$18	\$20, \$4
H_2	\$1, \$15	\$4, \$20	\$5, \$5

Your experiment is a simultaneous game with each player picking their actions simultaneously and independently. You observe the outcomes of the games and ever player i in the sample pick the action M_i.

2.4 Does the observed behavior reject a game theoretic explanation? Explain your answer.

Answer: No, we do not know preferences and hence payoffs of the players

2.5 Clearly define and explain the idea of a normal form of a game.

Answer: Notes

2.6 Postulate an explanation for the behavior observed in the experiment and create a normal form of this game with the unique Nash equilibrium of $\{M_1, M_2\}$.

Answer: any example with preferences ordered so that $\{M_1, M_2\}$ *is the unique NE.*

2.7 Can we use empirical data from an experiment of this form to test if the players are rational? Explain.

Answer: No, need preferences of players.